

DIT-IR Infrared Interface Module

Installation and operating instructions



DIT-IR Infrared Interface Module

Declaration of conformity	4
English (GB)	
Installation and operating instructions	5
Deutsch (DE)	
Montage- und Betriebsanleitung	12
Español (ES)	
Instrucciones de instalación y funcionamiento.	19
Français (FR)	
Notice d'installation et de fonctionnement	26
Italiano (IT)	
Istruzioni di installazione e funzionamento.	33
Nederlands (NL)	
Installatie- en bedieningsinstructies	40

Declaration of conformity

GB Declaration of Conformity

We, Grundfos, declare under our sole responsibility that the products DIT-L, DIT-M and DIT-IR, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

- EMC Directive (2004/108/EC).
Standard used: EN 61326-1: 2006, Class B
(DIT-IR: Interference resistance Class B).

ES Declaración de Conformidad

Nosotros, Grundfos, declaramos bajo nuestra entera responsabilidad que los productos DIT-L, DIT-M y DIT-IR, a los cuales se refiere esta declaración, están conformes con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

- Directiva EMC (2004/108/CE).
Norma aplicada: EN 61326-1: 2006, clase B
(DIT-IR: Resistencia de interferencia clase B).

IT Dichiarazione di Conformità

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti DIT-L, DIT-M e DIT-IR, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

- Direttiva EMC (2004/108/CE).
Norma applicata: EN 61326-1: 2006, Classe B
(DIT-IR: Resistenza alle interferenze Classe B).

DE Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte DIT-L, DIT-M und DIT-IR, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen:

- EMV-Richtlinie (2004/108/EG).
Norm, die verwendet wurde: EN 61326-1: 2006, Klasse B
(DIT-IR: Störfestigkeit Klasse B)

FR Déclaration de Conformité

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits DIT-L, DIT-M et DIT-IR, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

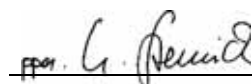
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).
Norme utilisée : EN 61326-1: 2006, Classe B
(DIT-IR: Immunité au brouillage Classe B).

NL Overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten DIT-L, DIT-M en DIT-IR waarop deze verklaring betrekking heeft, in overeenstemming zijn met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG Lidstaten betreffende:

- EMC Richtlijn (2004/108/EC).
Gebruikte norm: EN 61326-1: 2006, Klasse B
(DIT-IR: Weerstand tegen storingen Klasse B).

Pfintzal, 1st December 2010



Ulrich Stemick
Technical Director
Grundfos Water Treatment GmbH
Reetzstr. 85, D-76327 Pfintzal, Germany

Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

CONTENTS

	Page
1. Symbols used in this document	5
2. General information	5
2.1 Delivery contents	5
2.2 Application	5
2.3 Warranty	5
2.4 Avoidance of danger	5
2.5 Nameplate	6
3. Construction and function	6
3.1 System overview	6
3.2 Installation and operating instructions	6
3.3 Keypad overview	6
3.4 LED indicators	6
3.5 DIT-IR power supply	6
3.6 DIT-IR connection	7
4. Installation of the device driver and data acquisition software	7
4.1 Driver installation	7
4.2 Software installation	7
5. Start the application	8
5.1 Data transmission	8
5.2 Serial interface mode	8
5.3 PC interface mode	9
6. Disposal	11

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

These installation and operating instructions contain all information important for users of the DIT-IR infrared interface module.

Should you require further information, or should you encounter problems that are not described in sufficient depth in this manual, please contact Grundfos Water Treatment.

2.1 Delivery contents

Carefully inspect all items to ensure that every part of the list below is present and no visible damage has occurred during shipment.

If there is any damage or missing items, please contact your local distributor immediately.

A DIT-IR standard delivery comprises:

- 1 DIT-IR module in a plastic case
- 4 batteries (type AA/LR6)
- 1 USB cable
- 1 screwdriver with clip, red
- 1 installation and operating instructions
- 1 certificate of compliance.

2.2 Application

The DIT-IR infrared interface module is suitable for transferring data from a DIT-L or DIT-M photometer to a PC.

2.3 Warranty

A guarantee claim in accordance with our general terms of sale and delivery is only valid if the following requirements are fulfilled:

- The module has been used in accordance with the information in this manual.
- The module has not been dismantled or handled incorrectly.
- Repairs have only been carried out by authorised personnel.
- Only original parts have been used during repairs.

2.4 Avoidance of danger

**Warning**

Do not open or dismantle the module!

Cleaning, maintenance and repair must only be carried out by authorised personnel!

If safe operation cannot be ensured any longer, the DIT-IR infrared interface module is to be taken out of service and safeguarded against unintentional operation. This is the case under the following circumstances:

- The DIT-IR module has visible damage.
- The DIT-IR module seems to be inoperative.
- After a long storage time under unfavourable circumstances.

2.5 Nameplate

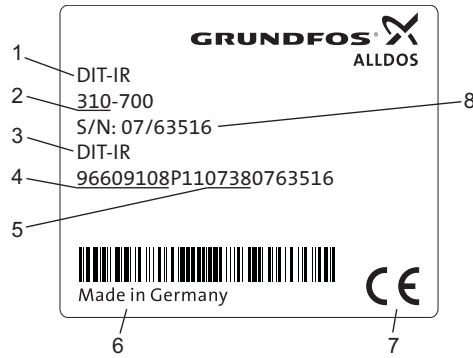


Fig. 1 Nameplate of DIT-IR

Pos.	Description
1	Type designation
2	Model
3	Product name
4	Product number
5	Year and week of production
6	Country of origin
7	Marks of approval, CE mark etc.
8	Serial number

3. Construction and function

3.1 System overview

The DIT-IR Infrared Interface Module (IRiM) receives data from a DIT-M or a DIT-L photometer and transmits them to one of the following interfaces:

- USB COM (PC)
- RS 232 connection (serial interface). All ASCII printers with serial interface can be used.

All interfaces can be connected at start-up. The interfaces can be selected by pressing the [Select] key of the DIT-IR. At start-up, the last selected interface is active.

3.1.1 System requirement for software installation

See also 4.2 Software installation.

Processor	Pentium 4/M or equivalent
RAM	512 MB
Screen resolution	Min. 1280 x 1024 pixels
Operating system	Windows XP
Disk space	90 MB

3.2 Installation and operating instructions

These installation and operating instructions serve as a reference for DIT-IR users and also for device firmware updates. The user must have prior knowledge of using Windows based applications.

3.3 Keypad overview

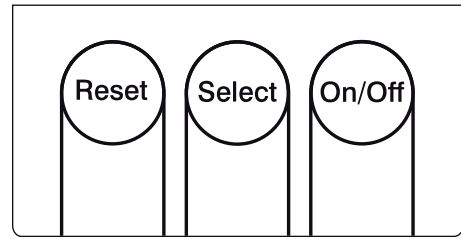


Fig. 2 DIT-IR keys

On/Off	Switches the module on and off
Select	For switching between the interface modes (scroll function)
Reset	Resets the DIT-IR board to default configuration

3.4 LED indicators

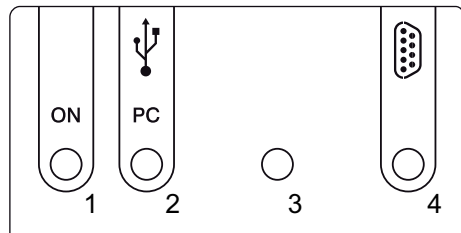


Fig. 3 DIT-IR keypad layout

1	Power indicator will be illuminated if the DIT-IR is running.
2	PC interface indicator will be illuminated, if the PC interface mode is selected, and the cable is connected with the PC. The LED flashes if the PC is not correctly connected.
3	USB interface and LED are not assigned.
4	Serial printer interface indicator will be illuminated, if the RS232 printer interface mode is selected. It will not check for the presence of a serial printer

3.5 DIT-IR power supply

The DIT-IR is powered by two optional sources:

- the USB-PC interface, if available
- 4 AA batteries, if the USB host power is not available.

If the DIT-IR is connected to the USB-PC interface, the batteries are not used. If the software is running, the power LED is illuminated. See fig. 3.

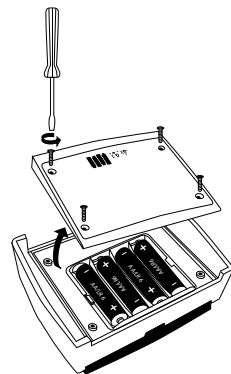


Fig. 4 DIT-IR battery compartment

Note

EC guideline 2006/66/EC requires users to return all used and worn-out batteries and accumulators. They must not be disposed of together with ordinary domestic waste.

3.6 DIT-IR connection

Place the DIT-IR as close as possible to the photometer.

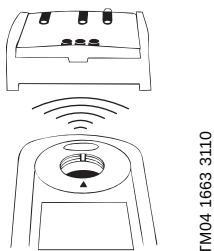


Fig. 5 DIT-IR connection

4. Installation of the device driver and data acquisition software

4.1 Driver installation

Note

Installation as described in this manual is based on a Windows XP platform. If you use another Windows operation system, please proceed analogously.

After the DIT-IR is connected to the USB interface for the first time, an installation assistant opens automatically. Make the selection shown below and click [Next].



Find the driver under "[CD drive name]:\inf" on the CD supplied. After inserting the CD and setting the following selection, start searching the CD by clicking [Next]:



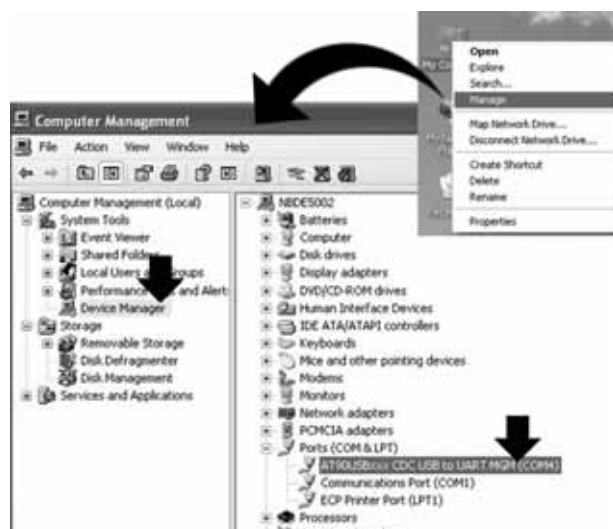
When the driver has been found, the message shown below appears. Confirm the installation by clicking [Continue Anyway].



The driver will be installed.

Complete the installation to close the installation assistant. Restart the PC.

A virtual COM port has been created. The address of the port can be seen in the device manager. It can be opened from the desktop as follows:



4.2 Software installation

The software for data acquisition and parameter setting can be found on the CD supplied.

Path: [CD drive name]:\Installer\Volume\setup.exe

Double-click to start the setup.exe file.



TM04 8134 3110

TM04 8132 3110

TM04 8135 3110

TM04 8133 3110

TM04 8136 3110

After starting the setup.exe file, the installation folder can be selected. It is recommended to confirm the suggested default folder by clicking [Next].



TM04 8137 3110

After confirmation, the software license agreement appears, which must be accepted.

Find other languages on the CD under "[CD drive name]\Installer\Volume\setup.exe".

To continue, click [Next].



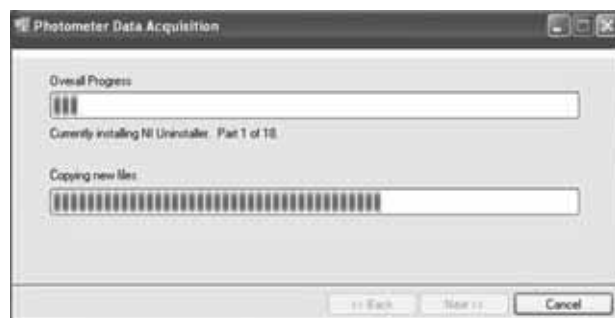
TM04 8138 3110

A summary of the programs to install appears. To start the installation, click [Next].



TM04 8139 3110

The software will be installed.



TM04 8140 3110

Complete the installation according to the menu navigation.

After the installation is completed, a restart is necessary. After the restart, the software can be used.

5. Start the application

Infrared data from the photometer will be received by DIT-IR at 9600 baud. The baud rate can be changed by the supplied software. See section 5.3.3 *Configuration of the DIT-IR settings*. When the unit is switched on, the last selected interface will be active.

5.1 Data transmission

Position the activated DIT-IR and the transmitting instrument, such as a photometer, in line of sight, face to face. See also fig. 5). Start print command at the transmitting instrument. The data will be sent to the DIT-IR module.

Caution

If the printer is not connected, the incoming data will be lost!

Note

The DIT-IR does not support and will not indicate any printer-related errors (paper jam, paper tray empty, etc.).

5.2 Serial interface mode

To scroll until the serial interface mode is set, press the [Select] key. The LED in front of the DB9 connector will flash up to indicate the serial interface mode. All ASCII printers with serial interface can be used.

5.2.1 Connection to a serial printer

Connect the serial printer to the DB9 connection of the DIT-IR module. Use a cable with non-transposed conductors.

Pin assignment of the serial cable:

- Pole 2 is connected to pole 2 of the other DB9 connection
- Pole 3 is connected to pole 3 of the other DB9 connection
- GND (pole 5 to pole 5).

5.2.2 Interface parameter

Set the same baud rate on the photometer (see the photometer installation and operating instructions), the DIT-IR (see 5.3.3 *Configuration of the DIT-IR settings*) and the serial printer (see manual).

5.3 PC interface mode

To scroll until the PC interface is set, press the [Select] key.

5.3.1 Connection to a PC

Connect the PC and the DIT-IR module port with a USB cable.

5.3.2 Data transmission to a PC

See also section 5.1 *Data transmission*

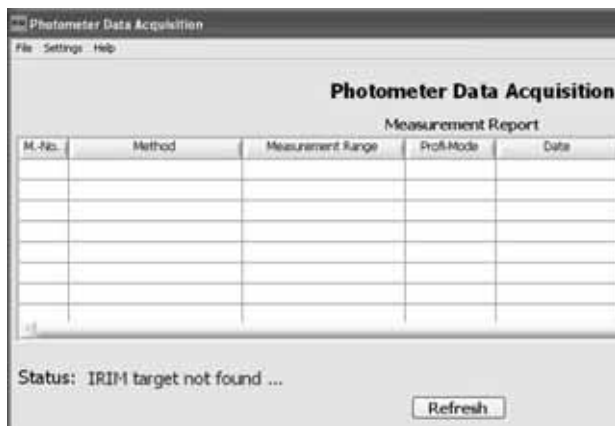
After software installation (section 4.2 *Software installation*), a folder containing a software shortcut will automatically be generated on the desktop.

After the software has been started, the following option window appears. A user name can be entered, the required language (German/English) can be chosen and the type of data report required can be selected.



TM04 8125 3110

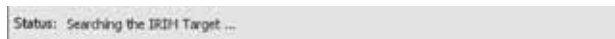
If no DIT-IR is connected when the software is started, the status message shown below appears after clicking [OK].



TM04 8126 3110

Connect the DIT-IR, select the appropriate interface and click [Refresh].

The following status message appears:

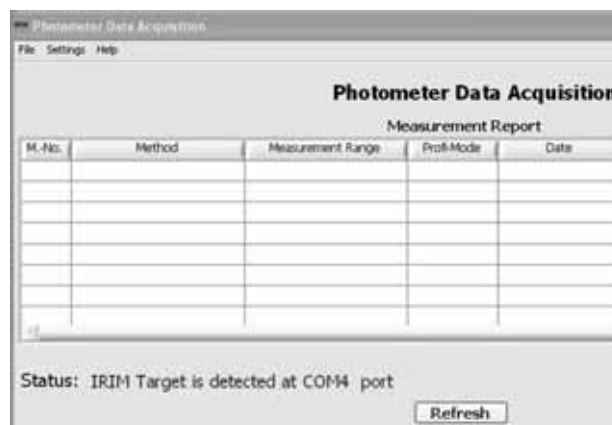


TM04 8127 3110

Option Excel

If the DIT-IR has been found, a window opens. Enter the file location and name.

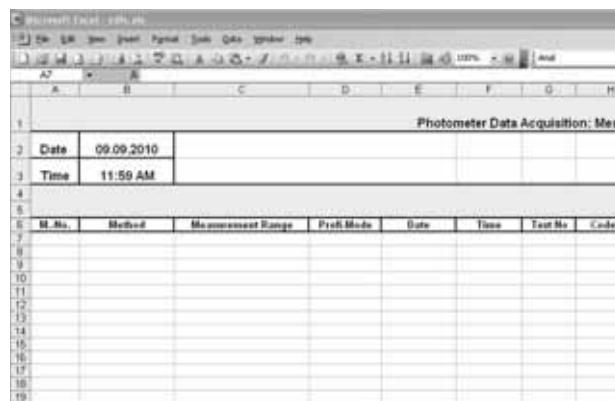
After confirming the location, the operating interface of the software opens:



TM04 8128 3110

In this window, the transmitted data can be viewed and stored. DIT-IR settings can be changed. See section 5.3.3 *Configuration of the DIT-IR settings*.

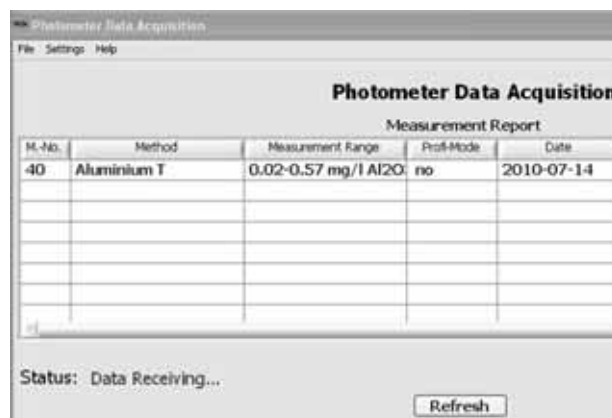
Simultaneously, an Excel template (result template) opens automatically, using the chosen language. The transmitted data will be stored in this sheet.



TM04 8129 3110

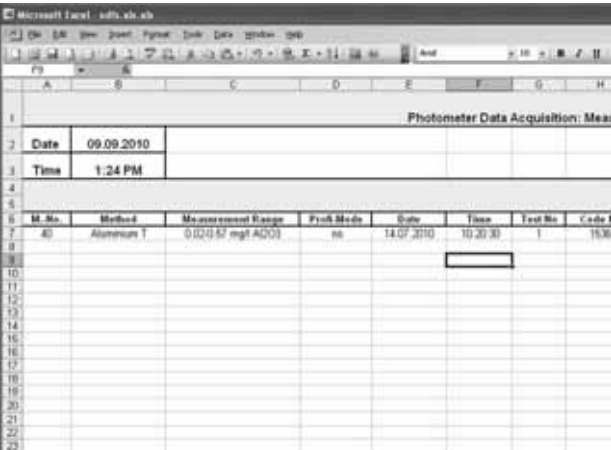
If the print command on the photometer has been pressed, the transmitted data can be seen in the operating interface as well as in the Excel template. The data will be saved automatically to the assigned location under the chosen name.

Data acquisition operating interface:



TM04 8178 3110

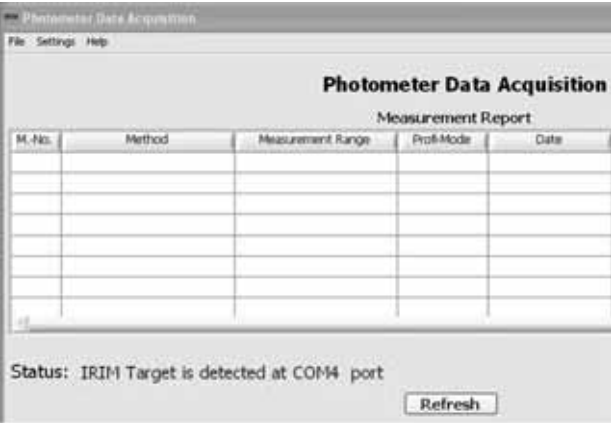
Data acquisition, Excel template:



TM04 8179 3110

Option, note pad

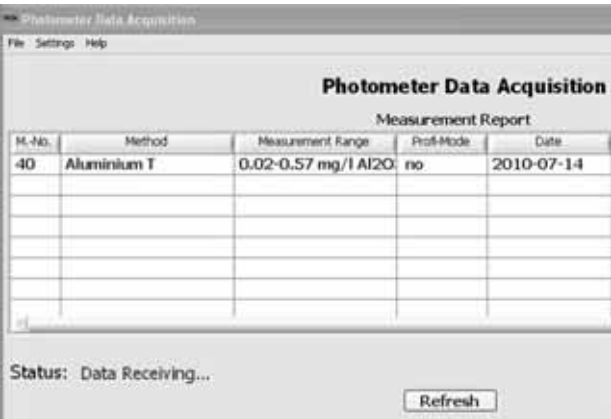
After selecting the note pad mode, a window opens. Enter the file location and name of the data to be transmitted.
After confirmation of the file location, the operating interface opens:



TM04 8128 3110

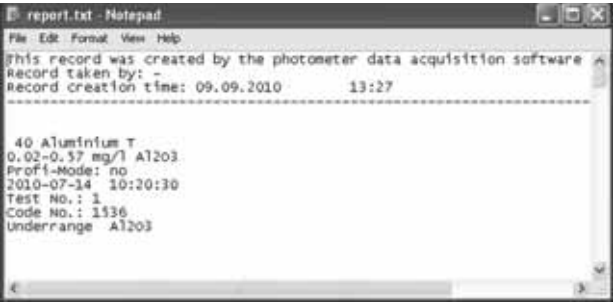
In this window, the transmitted data can be viewed and stored. Settings of the DIT-IR can be changed. See section 5.3.3 Configuration of the DIT-IR settings.

If the print command on the photometer has been pressed, the transmitted data can be seen in the operating interface.
Data acquisition operating interface:



TM04 8178 3110

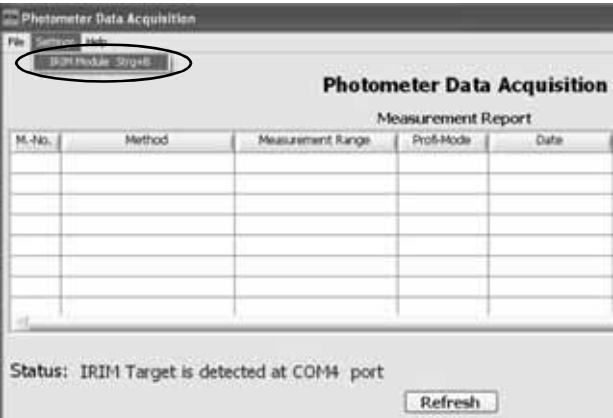
The data will be stored automatically in the assigned location and named as a .txt file.



TM04 8180 3610

5.3.3 Configuration of the DIT-IR settings

With the data acquisition software, various settings on the DIT-IR can be changed. The settings can be found under Settings > IRIM Module.

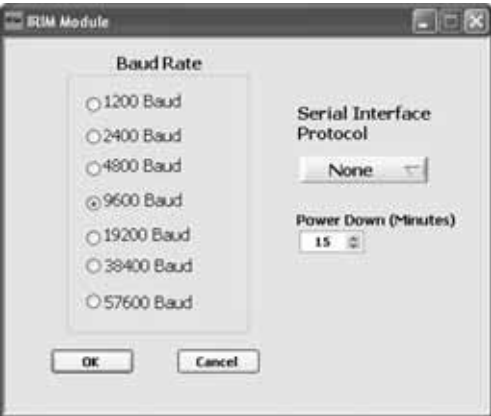


TM04 8181 3510

The window below opens. Set the baud rate, turn-off time ("Power Down") and the transmission protocol.

Default configuration: 9600 baud and protocol "None".

For any changes to the default settings, refer to the PC or device manuals. The turn-off time can be set between 1 and 120 minutes. The turn-off time is inactive if the DIT-IR is powered by the USB-PC connection.



TM04 8182 3110

6. Disposal

This product and all its associated parts must be disposed of in an environmentally friendly manner.

1. Use appropriate waste collection services.
2. If there is no such facility, or the facility refuses to accept the materials used in the product, the product can be sent to the nearest Grundfos company or service centre.

The 2006/66/EC guideline requires users to return all used and worn-out batteries and accumulators. They must not be disposed of in normal domestic waste.

Note

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote 34A
1619 - Garín
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belorussia

Представительство ГРУНДФОС в Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: +37517) 233 97 65
Факс: (37517) 233 9769
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco,
630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztocna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

Grundfos Alldos
Dosing & Disinfection
ALLDOS (Shanghai) Water Technology Co.
Ltd.
West Unit, 1 Floor, No. 2 Building (T 4-2)
278 Jinhu Road, Jin Qiao Export Processing
Zone
Pudong New Area
Shanghai, 201206
Phone: +86 21 5055 1012
Telefax: +86 21 5032 0596
E-mail: grundfosalldos-CN@grundfos.com

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
22 Floor, Xin Hua Lian Building
755-775 Huai Hai Rd, (M)
Shanghai 200020
PRC
Phone: +86-512-67 61 11 80
Telefax: +86-512-67 61 81 67

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čapkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS Water Treatment GmbH
Reetzstraße 85
D-76327 Pfintal (Söllingen)
Tel.: +49 7240 61-0
Telefax: +49 7240 61-177
E-mail: gwt@grundfos.com

Germany

GRUNDFOS GMBH
Schlüterstr. 33
D-40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
E-mail: infoservice@grundfos.de
Service in Deutschland:
E-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg. 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo,
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

México

Bombas GRUNDFOS de México S.A. de C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

România

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная 39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Grundfos (PTY) Ltd.
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentecilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
(Box 333) Lunnagårdsgatan 6
431 24 Mölndal
Tel.: +46(0)771-32 23 00
Telefax: +46(0)31-331 94 60

Switzerland

GRUNDFOS ALLDOS International AG
Schönmattdstraße 4
CH-4153 Reinach
Tel.: +41-61-717 5555
Telefax: +41-61-717 5500
E-mail: grundfosalldos-CH@grundfos.com

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloeam Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс.: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Представительство ГРУНДФОС в
Ташкенте
700000 Ташкент ул.Усмана Носира 1-й
туник 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

15.730231 V1.0
95727442 0111

ECM: 1069508

The name Grundfos, the Grundfos logo, and the payoff Be-Think-Innovate are registered trademarks owned by Grundfos Management A/S or Grundfos A/S, Denmark. All rights reserved worldwide.